

From Episodes to Abstractions: Latent Hierarchical Memory in 1,908 AI Conversations

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Abstract. We investigate whether AI conversation archives—externalized records of knowledge exploration through dialogue with large language models—contain hierarchical structure analogous to human semantic memory. From a longitudinal archive of 1,908 ChatGPT conversations spanning 29 months, we extract semantic concepts using LLM-based analysis, embed them in a 768-dimensional semantic space, and apply hierarchical agglomerative clustering to organize them into a four-level hierarchy: 500 *meta-concepts* $\rightarrow$ 50 *themes* $\rightarrow$ 8 *knowledge domains*. The concept co-occurrence network is small-world ($\sigma \approx 6.6$, validated against 100 Erdős-Rényi random graphs), matching the topology reported for human semantic memory networks. Meta-concept vocabulary follows Heaps’ law ($\beta = 0.320$), with growth dynamics distinguishable from random clustering ($p < 0.001$), indicating genuine categorical structure. The bipartite graph further reveals that 77% of episodes span multiple knowledge domains through shared concepts, and that cross-domain concept sharing is concentrated in specific domain pairs rather than distributed uniformly.

Keywords: semantic memory, hierarchical clustering, knowledge networks, complementary learning systems, AI conversations, small-world networks

1 Introduction

Over the past three years, millions of users have engaged in sustained dialogue with large language models (LLMs), producing conversation archives that constitute an unprecedented record of externalized knowledge exploration [23]. A single user’s ChatGPT history may contain thousands of episodes spanning programming, mathematics, philosophy, and creative writing—each episode a trace of a specific learning or problem-solving encounter. Yet these archives are typically treated as flat, chronological lists, discarding whatever structure might exist in the knowledge they collectively represent.

We ask: *does hierarchical structure emerge from these conversation archives when viewed through the lens of cognitive memory theory?*

Complementary Learning Systems (CLS) theory [13,11] provides a principled framework. CLS posits two interacting memory systems: a *hippocampal* system that rapidly encodes individual episodes, and a *neocortical* system that gradually extracts statistical regularities, forming abstract semantic knowledge. A prediction of this process is that novel experiences increasingly map to existing categories rather than creating new ones—a phenomenon measurable through vocabulary growth dynamics.

Independently, research on human semantic memory networks has established that conceptual knowledge organizes into *small-world* networks [18,17]: high local clustering with short global path lengths. Steyvers and Tenenbaum [18] showed this topology holds across Roget’s Thesaurus, WordNet, and word association norms [3], with clustering-to-path-length ratios corresponding to σ values of 5.6–15.3 under the later Humphries–Gurney formulation [9].

We apply these cognitive frameworks to a longitudinal archive of 1,908 ChatGPT conversations spanning December 2022 through April 2025. Our approach is bottom-up: we use an LLM to extract fine-grained semantic concepts from each episode, embed them in a shared vector space, and allow hierarchical structure to emerge through geometric clustering. The central contributions are:

1. A bipartite episode–concept graph and four-level hierarchy (1,908 episodes $\rightarrow$ 500 meta-concepts $\rightarrow$ 50 themes $\rightarrow$ 8 domains) with interpretable knowledge categories at each level;
2. Sublinear vocabulary growth (Heaps’ law $\beta = 0.320$), with growth dynamics distinguishable from a random-clustering null model ($p < 0.001$), showing that semantic structure creates meaningful categorical distinctions;
3. Small-world topology ($\sigma \approx 6.6$, against 100 Erdős–Rényi random graphs) in the concept co-occurrence network, consistent with human semantic memory benchmarks;
4. Many-to-many episode–concept associations, where 77% of episodes span multiple knowledge domains—structure invisible to partition-based methods;
5. Size-normalized asymmetric concept sharing between domains, revealing genuine semantic isolation punctuated by specific cross-domain dependencies.

2 Related Work

Complementary Learning Systems. CLS theory [13,11] explains how hippocampal episodic encoding and neocortical consolidation interact. The hippocampal system rapidly encodes individual experiences with pattern separation, while the neocortical system gradually extracts statistical regularities through interleaved replay. A key prediction is that novel experiences increasingly map to existing semantic categories—a form of vocabulary saturation that can be measured through growth dynamics like Heaps’ law [8]. Heaps’ law ($V(n) = K \cdot n^\beta$) was originally formulated for natural language vocabulary growth, where $\beta \approx 0.4$ – 0.6 [12]; lower β indicates stronger consolidation.

Cognitive Network Science. Semantic memory networks exhibit small-world topology [22]—high local clustering with short global path lengths [18,17]. Collins and Loftus [2] proposed spreading activation as the retrieval mechanism; Rosch [16] established hierarchical categorization. De Deyne et al. [3] confirmed small-world properties across 12,000 cues in a word association network. The small-world coefficient $\sigma = (C/C_{\text{rand}})/(L/L_{\text{rand}})$, as formalized by Humphries and Gurney [9], compares observed clustering and path lengths to Erdős-Rényi random graphs with matched node and edge counts. From the clustering and path length ratios reported by Steyvers and Tenenbaum [18], one can compute σ values of 5.6–15.3 across their three semantic networks, supporting their argument that small-world structure is a universal property of semantic memory.

Topic Modeling and Hierarchical Theme Discovery. LDA [1] discovers latent themes from word co-occurrence statistics; BERTopic [7] extends this with transformer embeddings and hierarchical topic merging. Hierarchical Dirichlet Processes [19] learn topic hierarchies nonparametrically. These approaches discover latent themes but do not produce named, interpretable concept nodes linked to specific episodes in a bipartite graph that supports downstream network analysis.

LLM-Based Knowledge Extraction. Recent work uses LLMs for ontology construction [6] and knowledge graph construction [15]. Graph RAG [4] uses LLM community summarization for retrieval-augmented generation. These systems extract structured knowledge from text but focus on downstream retrieval rather than analyzing the *structure* of the extracted knowledge itself.

Conversation Corpus Analysis. Large-scale LLM conversation datasets include WildChat [23] (1M ChatGPT interactions), LMSYS-Chat-1M [24] (1M multi-model conversations), and OpenAssistant [10] (161K crowd-sourced messages). These corpora support behavioral analysis and model evaluation but do not examine the semantic structure of individual user archives over time. Our prior work [20] constructed a semantic similarity network from the same archive using Louvain partitioning [5]. The present work differs fundamentally: extracting explicit concepts produces a bipartite graph capturing many-to-many associations that partition-based methods cannot represent.

3 Data and Methods

3.1 Dataset

The dataset comprises 1,908 ChatGPT conversations from a single user’s archive, spanning December 2022 through April 2025 (29 months). Conversations cover statistical methodology, software engineering, AI research, R programming, philosophy, and creative projects. The archive represents a naturalistic record of sustained AI-assisted knowledge exploration.

3.2 Concept Extraction

We extract semantic concepts from each episode using Claude Sonnet 3.5 v2¹ via the Anthropic API. For each episode, the model receives the first 20 messages of the conversation (each capped at 500 characters) and a structured prompt requesting 3–10 noun-phrase concepts (2–5 words each) capturing the specific knowledge explored. The prompt instructs: *“Be specific—use ‘Metropolis-Hastings algorithm’ rather than ‘statistics.’ Focus on what knowledge is explored, not how the conversation proceeds.”*

To process all 1,908 conversations efficiently, we deploy 20 parallel extraction agents, each operating independently on a partition of episodes. Because agents run in parallel, they cannot share a running vocabulary; post-hoc deduplication is performed geometrically via hierarchical clustering (§3.4).

3.3 Concept Embedding

Each unique concept is embedded using the `nomic-embed-text` model [14], producing 768-dimensional vectors via contrastive learning on a BERT-derived architecture. Embeddings are L2-normalized to enable cosine similarity comparisons.

3.4 Hierarchical Clustering

We apply hierarchical agglomerative clustering (Ward linkage [21], Euclidean distance) to the concept embedding matrix and cut the dendrogram at three heights:

- **Level 0:** 1,908 episodes (individual conversations)
- **Level 1:** 500 meta-concepts (semantic deduplication)
- **Level 2:** 50 themes (broad topic groups)
- **Level 3:** 8 domains (top-level knowledge areas)

The meta-concept level groups semantically equivalent concepts from parallel extraction agents (e.g., “bootstrap confidence intervals” and “BCa bootstrap confidence intervals”).

3.5 Bipartite Graph and Co-occurrence Network

We construct a bipartite graph $G = (E \cup C, L)$ where E is the set of 1,908 episodes, C is the set of 500 meta-concepts, and $(e, c) \in L$ whenever episode e was assigned at least one raw concept belonging to meta-concept cluster c . This yields 7,153 bipartite edges (mean 3.75 meta-concepts per episode, compared to 3.96 raw concepts per episode). From this bipartite graph, we derive a weighted co-occurrence network on meta-concepts: two meta-concepts $c_i, c_j \in C$ are linked whenever they co-occur in at least one episode, with edge weight $w(c_i, c_j) = |\{e \in E : (e, c_i) \in L \wedge (e, c_j) \in L\}|$.

¹ Model ID: `claude-3-5-sonnet-20241022`, temperature = 0.

4 Results

4.1 Concept Extraction and Deduplication

Extraction yields 7,555 raw concept mentions (3.96 per episode) mapping to 3,517 unique concepts after case-insensitive deduplication. The raw distribution is long-tailed: 79.4% of concepts appear in only one episode. The 500-cluster meta-concept level consolidates this further: singletons drop to <1%, and the mean frequency rises from 2.1 to 14.3 episodes per meta-concept (Table 1, Fig. 1).

Table 1. Effect of meta-concept deduplication on bipartite graph density.

Metric	Raw concepts	Meta-concepts
Unique concepts	3,517	500
Singletons (%)	79.4	< 1
Mean episodes/concept	2.1	14.3
Episode pairs sharing concepts	9,304	165,000

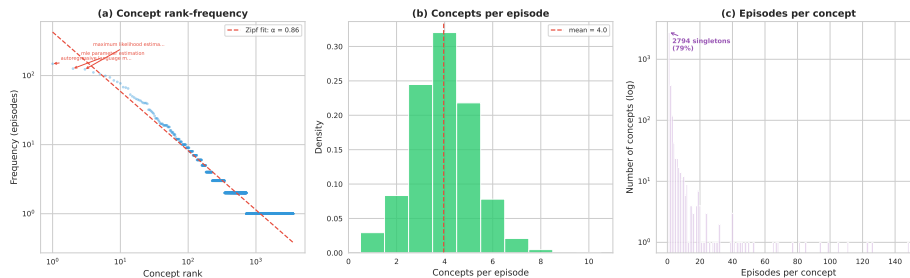


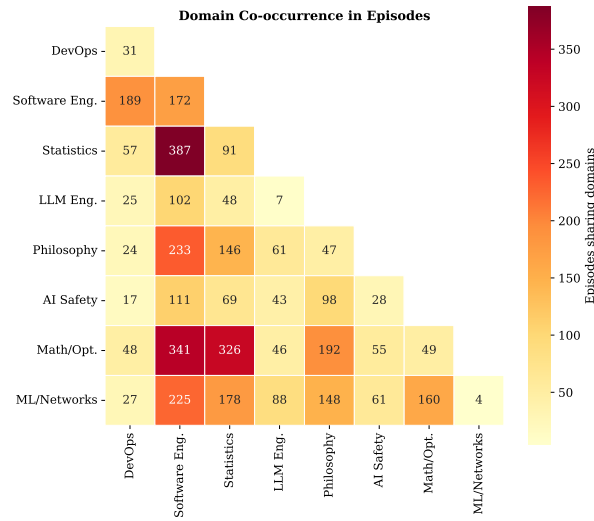
Fig. 1. Concept frequency distributions. **Left:** Zipf rank-frequency plot (Zipf exponent ≈ 0.86). **Center:** Concepts per episode (mean 3.96). **Right:** Episodes per concept showing the long tail (79% singletons).

4.2 Latent Hierarchy

Ward linkage at $k = 8$ produces eight interpretable knowledge domains (Table 2). The bipartite structure allows episodes to span multiple domains: only 22.5% are confined to a single domain; 42% span two, 25% span three, and 10% span four or more. Episodes spanning 3+ domains (35%) function as cross-domain integration points—e.g., “automatic differentiation for MLE in R” spans Software Engineering, Statistics, and Mathematics (Fig. 2).

Table 2. Eight knowledge domains from Ward linkage clustering.

Domain	Raw concepts	Representative concepts
Software Engineering	1,074	code refactoring, API design
Math & Optimization	634	simulation, Fisher info.
Statistical Methods	510	MLE, bootstrap, Weibull
Philosophy & AI Theory	489	consciousness, complexity
ML & Network Science	274	language models, networks
DevOps & Integration	273	APIs, hosting, admin
AI Safety & Research	177	alignment, evaluation
LLM Engineering	86	prompts, agents, tools

**Fig. 2.** Domain co-occurrence heatmap showing how many episodes span each pair of domains. Software Engineering co-occurs broadly; specialized pairs (Statistics–Math) show focused affinity.

4.3 Vocabulary Growth and Null Model

Processing episodes in alphabetical order by ID, the meta-concept vocabulary grows sublinearly following Heaps’ law $V(n) = K \cdot n^\beta$ with $\beta = 0.320$ (Fig. 3). Under 100 random episode orderings, $\beta = 0.241 \pm 0.007$, confirming that sublinear growth is robust to ordering, though the specific β value depends on the sequence (alphabetical ordering inflates β due to topical autocorrelation among lexicographically adjacent episode IDs). The raw concept vocabulary (before clustering) grows nearly linearly ($\beta = 0.931$), confirming that sublinear growth arises from the clustering step, not the extraction process.

Since clustering at $k = 500$ imposes a vocabulary ceiling that guarantees sublinear growth regardless of semantic content, we test whether the observed β

reflects genuine semantic structure or is purely a ceiling artifact. Using the same alphabetical episode ordering, we construct a null model by randomly permuting concept-to-cluster assignments (preserving cluster sizes) and computing β for 1,000 permutations. The null distribution has $\beta_{\text{null}} = 0.268 \pm 0.006$, and the observed $\beta = 0.320$ lies above the entire null distribution ($p < 0.001$). Semantic clustering produces a *higher* β (less sublinear, closer to linear) than random assignment: the semantically coherent clusters create meaningful distinctions, so encountering a genuinely new meta-concept requires exploring novel intellectual territory rather than merely encountering new vocabulary.

Clustering sensitivity analysis confirms robustness: β increases monotonically from 0.061 ($k = 50$) to 0.467 ($k = 1,000$), but the qualitative finding of sublinear growth is stable across $k = 200$ –1,000 (Table 4).

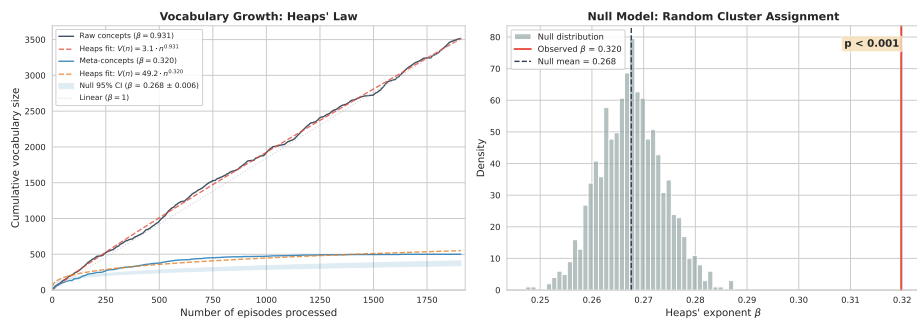


Fig. 3. Vocabulary growth and null model validation. **Left:** Cumulative vocabulary following Heaps’ law ($\beta = 0.931$ for raw concepts, $\beta = 0.320$ at the meta-concept level). The shaded band shows the 95% null model range under random cluster assignment. **Right:** Null model distribution—the observed $\beta = 0.320$ lies entirely above the null ($\beta_{\text{null}} = 0.268$, $p < 0.001$), confirming that semantic structure shapes the growth dynamics.

4.4 Small-World Topology

The meta-concept co-occurrence network (499 of 500 meta-concepts; one singleton has no co-occurrence edges; 5,935 edges) exhibits small-world properties (Table 3). Following the Humphries–Gurney formulation [9], we compare clustering and path length to 100 Erdős–Rényi $G(n, m)$ random graphs with matched node and edge counts. The clustering coefficient is $6.9\times$ higher than random, while the path length is only $1.05\times$ longer, yielding $\sigma \approx 6.6$. This is comparable to σ values derivable from the clustering and path-length ratios reported for human semantic memory networks [18,3]—Roget’s ($\sigma \approx 13$), WordNet ($\sigma \approx 15$), and word associations ($\sigma \approx 5.6$)—though direct comparison is approximate since those studies used different random graph baselines. Clustering sensitivity con-

firms robustness: σ ranges from 2.0 ($k = 200$) to 12.6 ($k = 700$), remaining above 1 across all tested values of k (Table 4).

Table 3. Small-world analysis of the concept co-occurrence network. Random graph baselines are means over 100 Erdős-Rényi $G(n, m)$ graphs.

Property	Value	Interpretation
Nodes	499	
Edges	5,935	
Clustering C	0.329	
C/C_{rand}	6.89	High local clustering
Path length L	2.37	
L/L_{rand}	1.05	Short global paths
Small-world σ	6.57	Strongly small-world
Mean degree	23.8	

Table 4. Clustering sensitivity: key metrics across meta-concept granularity k . Both β and σ vary with k but the qualitative findings (sublinear growth, small-world topology) are robust across $k = 200$ –1,000.

k	Heaps' β	σ	C	Edges
50	0.061	1.1	0.830	906
100	0.103	1.4	0.652	2,286
200	0.158	2.0	0.425	4,111
300	0.225	2.9	0.351	5,073
500	0.320	6.57	0.329	5,935
700	0.382	12.6	0.351	6,593
1000	0.467	25.9	0.401	7,291

4.5 Asymmetric Concept Flow

The bipartite graph enables a directed analysis that the undirected co-occurrence network cannot capture: for each pair of domains (A, B) , we measure the fraction of B 's episodes that contain at least one concept from domain A . We call this fraction the *broadcast reach* of domain A —the share of non- A episodes containing at least one of A 's concepts. Conversely, the *porosity* of domain A is the fraction of A 's episodes containing at least one foreign concept. If concept sharing were symmetric, broadcast reach from A to B would equal the reverse; in practice, it is asymmetric (Fig. 4).

Software Engineering has the highest broadcast reach (40% of all non-SE episodes), penetrating all seven other domains. However, Software Engineering

also contains 30.5% of all raw concepts (1,074 of 3,517), raising the question of whether its broad reach reflects genuine cross-domain dependency or a domain size effect. To control for this, we compare observed flow to a null model that randomly reassigns concept-domain labels (preserving domain sizes, 100 permutations). The observed/expected ratios fall below 1.0 for most domain pairs (mean = 0.61), confirming that the eight domains capture genuine semantic boundaries: concepts cluster within domains more tightly than expected under random concept-domain assignment.

Within this overall isolation, specific domain pairs show elevated sharing relative to the null. LLM Engineering $\rightarrow$ ML & Network Science has an observed/expected ratio of 2.67, and LLM Engineering $\rightarrow$ AI Safety reaches 1.27. ML & Network Science has the highest porosity (30% of its episodes contain foreign concepts), functioning as an interdisciplinary junction where concepts from multiple domains converge.

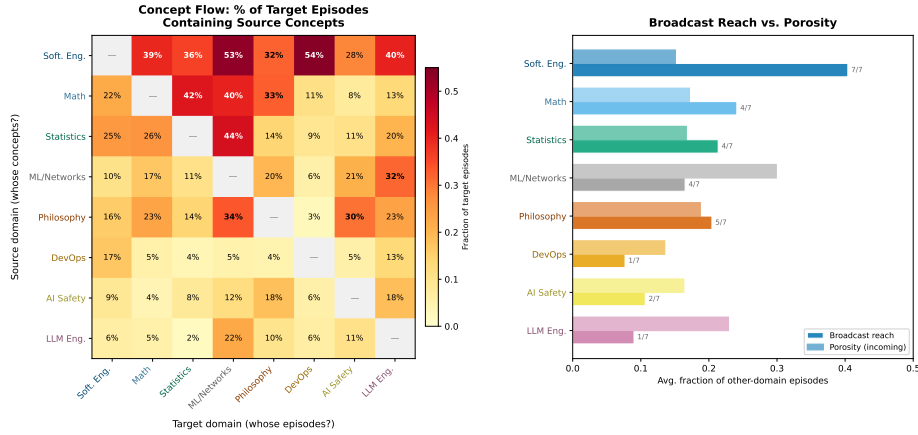


Fig. 4. Asymmetric concept flow between domains. **Left:** Flow matrix showing the fraction of target-domain episodes (columns) containing source-domain concepts (rows). The matrix is asymmetric: Software Engineering exports broadly (top row), while ML & Network Science absorbs from all domains (column 4). **Right:** Broadcast reach (fraction of foreign episodes containing a domain’s concepts) versus porosity (fraction of a domain’s episodes containing foreign concepts). Numbers indicate reach breadth (domains penetrated above 15%).

4.6 Bipartite Network Visualization

Figure 5 visualizes the full bipartite structure as a radial layout: meta-concepts occupy an inner ring, episodes an outer ring, both grouped by domain sector. Bipartite edges connect episodes to their constituent meta-concepts, while co-occurrence edges link concepts that share episodes. The visualization makes

the asymmetric flow visible: Software Engineering episodes (lower right) send concept links into every sector, while Philosophy episodes (lower left) connect almost exclusively to their own domain’s concepts.

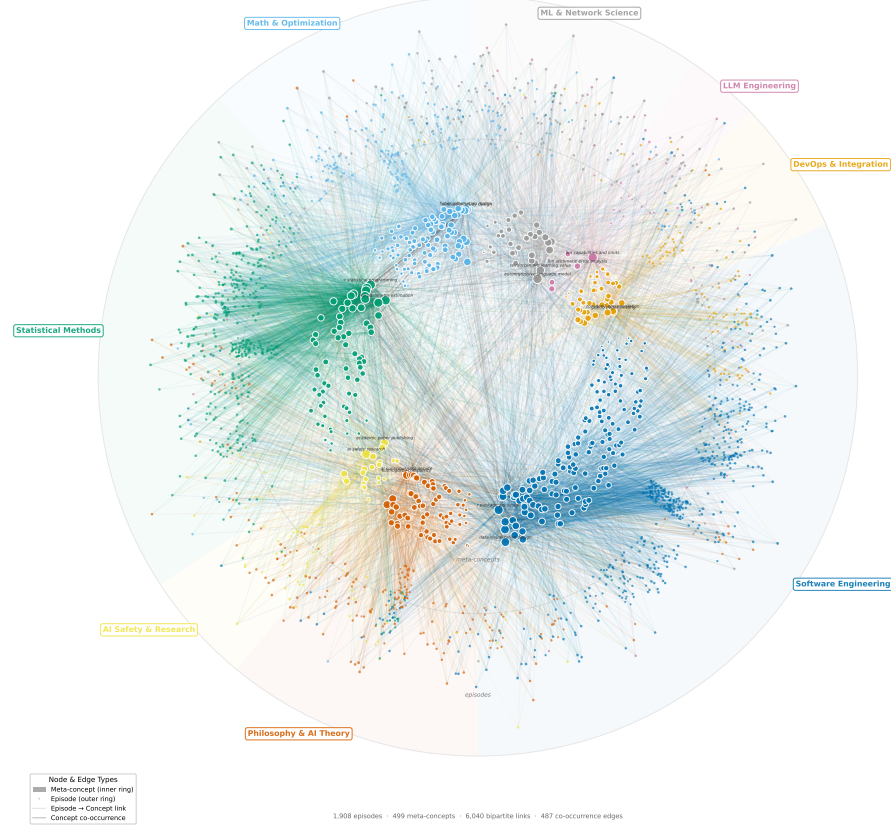


Fig. 5. Bipartite knowledge map. Inner ring: 499 meta-concepts (colored and sized by domain and frequency). Outer ring: 1,908 episodes (colored by primary domain). Radial lines show episode–concept associations (filtered to meta-concepts appearing in ≥ 8 episodes); gray lines show concept co-occurrence (≥ 4 shared episodes). Sector size is proportional to episode count. Cross-sector bipartite edges reveal the directed concept flow: foundational domains export concepts broadly, while applied domains absorb.

Figure 6 provides a complementary zoom-in view: two specific episodes from different domains—“AI Language Models Compared” (Statistical Methods) and “Language Models and Computation” (Philosophy & AI Theory)—share five

meta-concepts spanning four knowledge domains. These episodes approach the same intellectual territory (the nature of language model prediction) from fundamentally different disciplinary angles, yet the bipartite graph reveals their deep connection through shared concepts like Solomonoff induction, Kolmogorov complexity, and autoregressive language models. A partition-based method [5] would assign these episodes to separate communities; the bipartite concept hierarchy captures their relationship explicitly.

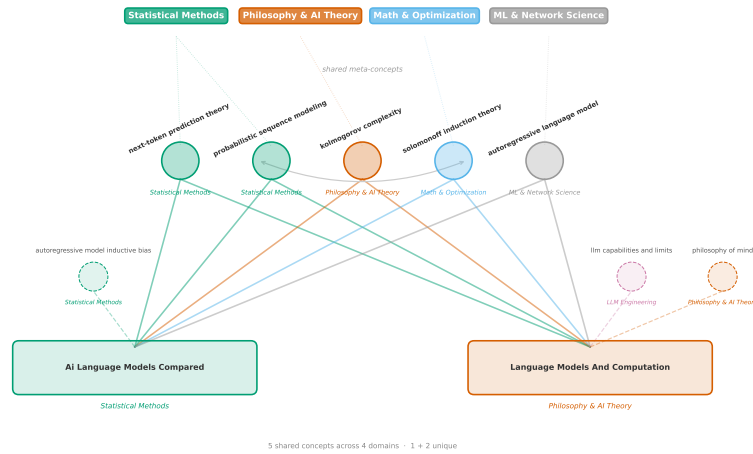


Fig. 6. Zoom-in: two episodes from different domains connected through shared concepts. “AI Language Models Compared” (Statistical Methods) and “Language Models and Computation” (Philosophy & AI Theory) share 5 meta-concepts spanning 4 domains (top row). Solid circles and lines show shared concepts; dashed circles show concepts unique to each episode. Each episode also has unique concepts reflecting its disciplinary focus: inductive bias (Statistics) vs. philosophy of mind and LLM capabilities (Philosophy/LLM Engineering).

5 Discussion

Vocabulary Growth and Semantic Structure. The meta-concept vocabulary grows sublinearly ($\beta = 0.320$), but since clustering at fixed $k = 500$ guarantees sublinear growth regardless of content, the absolute β value is not itself evidence of consolidation. The null model comparison reveals a more nuanced finding: semantic clustering produces *higher* β (less sublinear) than random cluster assignment ($\beta_{\text{null}} = 0.268$, $p < 0.001$). This means the semantically coherent clusters create meaningful categorical distinctions—encountering a genuinely new meta-concept

requires exploring novel intellectual territory, whereas random clusters generate “accidental” consolidation through arbitrary groupings. Under random assignment, conceptually unrelated items share clusters by chance, so any new episode is likely to fall into an existing cluster through accidental overlap; under semantic clustering, clusters are internally coherent and only genuinely related episodes share a meta-concept. CLS theory would predict that meaningful categories *facilitate* consolidation (lower β), but we observe the opposite: semantically coherent categories *resist* consolidation. This suggests that the archive’s categorical structure maintains discriminable knowledge representations—a property consistent with the role of semantic memory in preserving distinctions, even if it does not follow the consolidation-efficiency prediction of CLS.

Small-World Topology. The small-world coefficient $\sigma \approx 6.6$, validated against 100 Erdős-Rényi random graphs [9], matches human semantic memory benchmarks [18]. While we do not claim conversation archives *are* semantic memory, this structural similarity suggests they function as an externalized analog: concepts connected through co-occurrence in knowledge episodes, exhibiting the same balance of local coherence and global accessibility. The sensitivity analysis (Table 4) shows $\sigma > 1$ is robust across $k = 100$ –1,000, with the qualitative finding independent of the specific clustering granularity.

Many-to-Many Associations and Cross-Domain Flow. The bipartite graph captures knowledge polysemy: 77% of episodes span multiple domains, with 35% spanning three or more. Partition-based methods [5] cannot represent this multiplicity. The bipartite structure also reveals asymmetric concept flow between domains. After normalizing for domain size, most cross-domain flow is *below* what random concept-domain assignment would predict (mean observed/expected = 0.61), confirming that the domains represent genuine semantic clusters rather than arbitrary partitions. A few specific pairs show elevated flow (LLM Engineering $\rightarrow$ ML & Network Science at $2.67\times$ expected), suggesting targeted conceptual dependencies. For downstream applications like Graph RAG [4], the bipartite structure and domain hierarchy could enable concept-level indexing and retrieval.

Limitations. This is a single-user case study; different users would produce different hierarchies. Validating structural universality—whether sublinear growth and small-world topology are preserved across users—requires multi-user datasets. Hierarchy cut points ($k = 500, 50, 8$) are chosen, not discovered; the silhouette analysis shows generally increasing scores with k (monotonic for $k \geq 20$, with no distinct peak), indicating no natural cluster boundaries. Concept extraction depends on the specific LLM and prompt used; alternative models or prompts would produce different concept sets. Despite these limitations, the convergence with human semantic memory benchmarks across multiple metrics suggests that the structural properties may reflect general features of categorical knowledge systems rather than idiosyncrasies of this particular archive.

6 Conclusion

We have shown that 1,908 AI conversations contain latent hierarchical structure. LLM-extracted concepts organize into a four-level hierarchy with interpretable domains, sublinear vocabulary growth validated against a random-clustering null model ($p < 0.001$), and small-world topology ($\sigma \approx 6.6$, against 100 random graphs) matching human semantic memory benchmarks. The bipartite structure captures cross-domain integration that partition-based methods miss, while size-normalized flow analysis reveals genuine semantic isolation between domains punctuated by specific cross-domain conceptual dependencies. These results, presented as a single-user case study, position AI conversation archives as structured knowledge artifacts whose organization exhibits structural parallels—particularly small-world topology—with human semantic memory networks.

Data and Code Availability. Code for the analysis pipeline is available at <https://github.com/queelius/chatgpt-complex-net> (DOI: 10.5281/zenodo.15314235).

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